A GEOMETRIC INTRODUCTION TO THE REAL LOCAL LANGLANDS CORRESPONDENCE

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About these notes. This document was generated by Codex from my handwritten lecture notes prepared for the Yale Graduate Student Seminar. These notes are purely expository and contain no new mathematical content. It certainly contains mistakes; feel free to contact me if you find any. The explicit Langlands correspondence calculations for $\mathrm{PGL}_2(\mathbb{C})$ and $\mathrm{SL}_2(\mathbb{C})$ are taken from this article [2].

1. What groups? What representations?

1.1. **Real forms and inner classes.** Let G be a connected complex reductive algebraic group. The real reductive groups considered in this talk are its real forms.

Definition 1.1. A *real form* of G is an antiholomorphic group automorphism

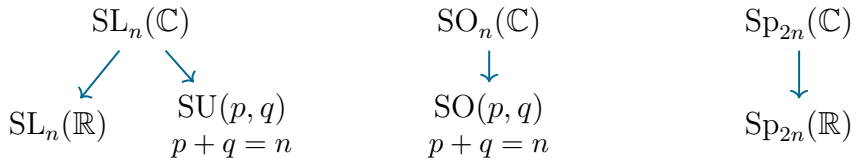
$$\sigma : G \longrightarrow G, \qquad \sigma^2 = \mathrm{id}\,.$$

Its group of real points is

$$G(\mathbb{R}, \sigma) = G^\sigma.$$

We write $G_{\mathbb{R}}$ when a particular real form has been fixed.

Here are some examples; the list is not intended to be exhaustive.



The complex group is fixed; changing the antiholomorphic involution changes the real group of fixed points.

Definition 1.2. Two real forms σ_1, σ_2 are *equivalent* if they are conjugate by an inner automorphism:

$$\sigma_1 = \mathrm{Ad}(g) \circ \sigma_2 \circ \mathrm{Ad}(g)^{-1} \qquad \text{for some } g \in G.$$

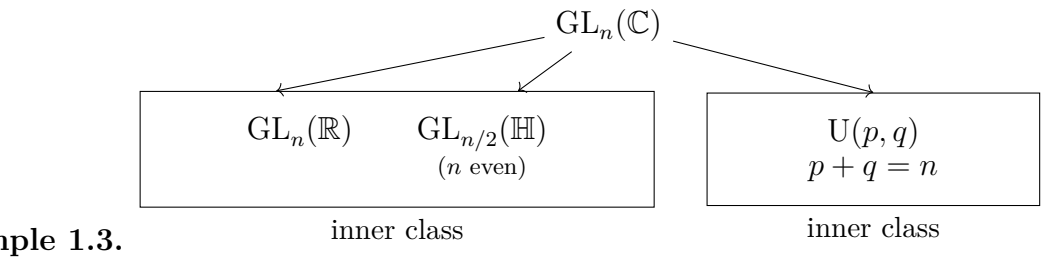
Two equivalence classes of real forms are *inner to each other* if their representatives σ_1, σ_2 satisfy

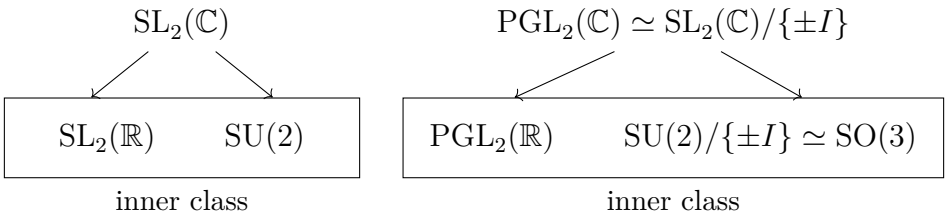
$$\sigma_1 = \mathrm{Ad}(g) \circ \sigma_2 \qquad \text{for some } g \in G.$$

This condition is independent of the chosen representatives and defines an equivalence relation on the set of equivalence classes of real forms. The classes for this relation are called *inner classes*.

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Example 1.3.





1.2. **Admissible representations.**

Definition 1.4. A *continuous representation* of $G_{\mathbb{R}}$ is a continuous action

$$\pi : G_{\mathbb{R}} \times V \longrightarrow V$$

by complex-linear automorphisms, where V is a Hausdorff, quasi-complete, locally convex complex topological vector space. A representation is *unitary* if V is a Hilbert space and the action preserves its Hermitian inner product.

Fix a maximal compact subgroup $K_{\mathbb{R}} \subset G_{\mathbb{R}}$.

Definition 1.5. The representation V is *admissible* if

$$\dim_{\mathbb{C}} \mathrm{Hom}_{K_{\mathbb{R}}}(\tau, V) < \infty \qquad \text{for every } \tau \in \widehat{K_{\mathbb{R}}}.$$

A vector $v \in V$ is $K_{\mathbb{R}}$-*finite* if the span of its $K_{\mathbb{R}}$-orbit is finite-dimensional.

Example 1.6. The standard action of $\mathrm{SL}_2(\mathbb{R})$ on $\mathbb{R}^2$ induces an action on

$$X = \mathbb{P}^1(\mathbb{R}) \simeq S^1/(z \sim -z).$$

The action on X gives a lot of representations on all kinds of function spaces:

$$\begin{matrix} C^\infty(X), & L^2(X) \\ \mathcal{S}(X), & C^{-\infty}(X) \end{matrix} \xrightarrow{K_{\mathbb{R}}\text{-finite}} \mathbb{C}[z^2, z^{-2}]$$

All of them are admissible. The action on X does not preserve the usual measure $\frac{1}{2\pi} d\theta$ (exercise); the action on square integrable half-densities supplies the unitary normalization.

Put

$$\mathfrak{g}_0 = \mathrm{Lie}(G_{\mathbb{R}}), \qquad \mathfrak{g} = \mathfrak{g}_0 \otimes_{\mathbb{R}} \mathbb{C}, \qquad K = (K_{\mathbb{R}})_{\mathbb{C}}.$$

Proposition 1.7. *If V is an admissible representation of $G_{\mathbb{R}}$, then the subspace of $K_{\mathbb{R}}$-finite vectors are smooth and dense in V.*

Proof. Firstly, we show that $V_{K\text{-fin}} \subseteq V^\infty$. We know that the subspace of smooth functions V^∞ is dense in V (Gårding's theorem). For each $\pi \in \hat{K}$, we can define the projection

operator $P_\pi : V \to V(\pi)$:

$$P_\pi = \int_{K_\mathbb{R}} \overline{\chi_\pi(k)}\pi(k)dk.$$

Quasi-completeness make sure that this integral exists. It is easy to verify that $P_\pi(V^\infty) \subseteq V^\infty$, and P_π is continuous. Therefore, $P_\pi(V^\infty)$ is dense in $V(\pi)$. Since $V(\pi)$ is finite-dimensional, we have $P_\pi(V^\infty) = V(\pi)$. Hence, the subspace of $K_\mathbb{R}$-finite vectors is dense in V.

Second, we show that $V_{K\text{-fin}} = \oplus_{\pi\in\hat{K}} V(\pi)$ (equality leave as an exercise) is dense in V. Let $R(K)$ denote the ring of all matrix coefficients of finite-dimensional representations of $K_\mathbb{R}$. For any $f \in R(K)$ and $v \in V$, we define

$$v_f = \int_{K_\mathbb{R}} f(k)\pi(k)vdk.$$

Then $\pi(k_0)v_f = v_{L_{k_0}f}$, since f is K-finite, we have $v_f \in V_{K\text{-fin}}$. Suppose that $V_{K\text{-fin}}$ is not dense in V, then by Hahn-Banach theorem there exists a nonzero continuous linear functional $\lambda \in V^*$ such that $\lambda(V_{K\text{-fin}}) = 0$. For any $v \in V$, consider the function $F_v(k) = \lambda(\pi(k)v)$. Then $F_v \in C(K_\mathbb{R})$ and for any $f \in R(K)$, $\int_{K_\mathbb{R}} f(k)\overline{F_v}(k)dk = 0$. Therefore, $F_v = 0$ by the Peter-Weyl theorem. In particular, $F_v(e) = \lambda(v) = 0$ for all $v \in V$, which contradicts the assumption that λ is nonzero. Hence, $V_{K\text{-fin}}$ is dense in V. □

Definition 1.8. A $(\mathfrak{g}, K)$-*module* is a complex vector space W with compatible $\mathfrak{g}$- and K-actions satisfying:

(1) the K-action is locally finite and algebraic;
(2) $k(Xw) = (\mathrm{Ad}(k)X)(kw)$ for $k \in K$, $X \in \mathfrak{g}$, and $w \in W$;
(3) the restriction of the $\mathfrak{g}$-action to $\mathrm{Lie}(K)$ is the differential of the K-action.

A *Harish-Chandra module* is a $(\mathfrak{g}, K)$-module of finite length; equivalently, it is admissible and finitely generated over $U(\mathfrak{g})$. Write $\mathcal{HC}(\mathfrak{g}, K)$ for this category.

Two admissible representations are *infinitesimally equivalent* if their underlying $(\mathfrak{g}, K)$-modules are isomorphic. Thus passage to K-finite smooth vectors gives the algebraic description of the representations relevant here.

$$\frac{\left\{\begin{array}{c}\text{Irreducible admissible}\\ G_\mathbb{R}\text{-representations}\end{array}\right\}}{\text{infinitesimal equivalence}} \overset{\text{1:1}}{\longleftrightarrow} \frac{\left\{\begin{array}{c}\text{Irreducible}\\ \text{Harish-Chandra modules}\end{array}\right\}}{\text{isomorphism}}$$

$$[V] \longmapsto \left[V_{K\text{-fin}}\right]$$

Theorem 1.9 (Harish-Chandra)**.** *Every irreducible unitary representation of $G_\mathbb{R}$ is admissible. Passage to $K_\mathbb{R}$-finite vectors gives a bijection between irreducible unitary representations, up to unitary equivalence, and irreducible unitarizable Harish-Chandra modules, up to isomorphism.*

$$\frac{\left\{\begin{matrix}\text{Irreducible unitary}\\ G_{\mathbb{R}}\text{-representations}\end{matrix}\right\}}{\text{unitary equivalence}} \overset{\text{1:1}}{\longleftrightarrow} \frac{\left\{\begin{matrix}\text{Irreducible unitarizable}\\ \text{Harish-Chandra modules}\end{matrix}\right\}}{\text{isomorphism}}$$

For classification, it is useful to keep in mind the inclusions

$$\widehat{G}_a \supseteq \widehat{G}_u.$$

These denote the irreducible admissible, unitary representations.

Two main problems in the representation theory of Lie groups are:

(1) **Classification:** classify irreducible representations up to equivalence.
(2) **Structure:** understand these irreducible representations—their realizations, characters, and how they occur as composition factors of standard representations.

In this talk, Section 3 addresses **classification** through the enhanced Langlands correspondence. Section 4 turns to **structure**, using geometric parameters and perverse sheaves to study representation-theoretic multiplicities.

2. Representations of $\mathrm{SL}_2(\mathbb{R})$ and $\mathrm{PGL}_2(\mathbb{R})$

2.1. **Infinitesimal characters.** Choose a Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$ with Weyl group W. Write $Z(\mathfrak{g}) = Z(U(\mathfrak{g}))$.

Theorem 2.1 (Harish-Chandra isomorphism)**.** *With the usual ρ-shift, there is an isomorphism*

$$Z(\mathfrak{g}) \simeq S(\mathfrak{h})^W, \qquad \operatorname{MaxSpec} Z(\mathfrak{g}) \simeq \mathfrak{h}^*/W.$$

For $\lambda \in \mathfrak{h}^*$, let $\chi_\lambda : Z(\mathfrak{g}) \to \mathbb{C}$ be the corresponding central character. A module V has *generalized infinitesimal character* λ if, for every $z \in Z(\mathfrak{g})$ and $v \in V$,

$$(z - \chi_\lambda(z))^N v = 0 \qquad \text{for } N \text{ sufficiently large.}$$

Write $\mathcal{HC}_\lambda(\mathfrak{g}, K)$ for the corresponding subcategory. An irreducible Harish-Chandra module has an ordinary infinitesimal character, by Schur's lemma. In the ρ-shifted convention,

$$\mathbf{1} \in \mathcal{HC}_\rho(\mathfrak{g}, K).$$

We now study the irreducible representations of $\mathrm{SL}_2(\mathbb{R})$ with infinitesimal character ρ.

2.2. **Parabolic induction.** Let

$$B_{\mathbb{R}} = \left\{ \begin{pmatrix} x & b \\ 0 & x^{-1} \end{pmatrix} : x \in \mathbb{R}^\times,\ b \in \mathbb{R} \right\} \subset \mathrm{SL}_2(\mathbb{R}).$$

For $\epsilon \in \{0, 1\}$ and $\nu \in \mathbb{C}$, define

$$\chi_{\epsilon,\nu}\left(\begin{pmatrix} x & b \\ 0 & x^{-1} \end{pmatrix} \right) = (\operatorname{sgn} x)^\epsilon |x|^\nu.$$

The normalized principal series is

$$\begin{aligned} P_{\epsilon,\nu} &= \operatorname{Ind}_{B_{\mathbb{R}}}^{\mathrm{SL}_2(\mathbb{R})} \chi_{\epsilon,\nu} \\ &= \{ f \in C^\infty(\mathrm{SL}_2(\mathbb{R})) : f(gb) = \delta_{B_{\mathbb{R}}}(b)^{1/2} \chi_{\epsilon,\nu}(b) f(g) \}_{K_{\mathbb{R}}\text{-fin}}, \end{aligned} \tag{2.1}$$

where $\delta_{B_{\mathbb{R}}}(b)^{1/2} = |x|$ and $G_{\mathbb{R}}$ acts by left translation. We use the same notation for the smooth globalization when no confusion is possible.

Equivalently, the smooth induced representation is the space of sections of the associated character line bundle twisted with half-densities:

$$\Gamma^{\infty}\left(G_{\mathbb{R}}/B_{\mathbb{R}}, (G_{\mathbb{R}} \times^{B_{\mathbb{R}}} \mathbb{C}_{\chi_{\epsilon,\nu}}) \otimes |\omega_{G_{\mathbb{R}}/B_{\mathbb{R}}}|^{1/2}\right).$$

Using the invariant pairing on $\mathfrak{sl}_2$,

$$T^*(G_{\mathbb{R}}/B_{\mathbb{R}}) \simeq G_{\mathbb{R}} \times^{B_{\mathbb{R}}} \mathfrak{n}_{\mathbb{R}}, \qquad |\omega_{G_{\mathbb{R}}/B_{\mathbb{R}}}|^{1/2} \simeq G_{\mathbb{R}} \times^{B_{\mathbb{R}}} \mathbb{C}_{|\wedge^{top}\mathfrak{n}|^{\frac{1}{2}}}.$$

At $\nu = 1$, $P_{1,1}$ is irreducible, while $P_{0,1}$ has the trivial representation as its irreducible quotient.

2.3. **Cohomological induction.** Take the compact Cartan subgroup

$$T_{\mathbb{R}} = \mathrm{SO}(2) = \left\{ \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} : \theta \in \mathbb{R} \right\}.$$

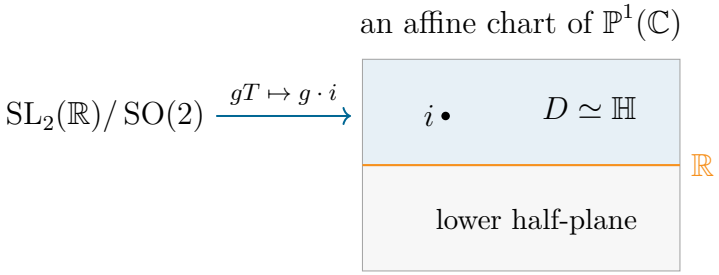
Its character group is $\widehat{T}_{\mathbb{R}} \simeq \mathbb{Z}$; denote the characters by χ_n. The homogeneous space

$$D = \mathrm{SL}_2(\mathbb{R})/T_{\mathbb{R}} \simeq \mathbb{H} \longhookrightarrow \mathrm{SL}_2(\mathbb{C})/B_{\mathbb{C}} \simeq \mathbb{P}^1(\mathbb{C})$$

is an open complex orbit, where $B_{\mathbb{C}}$ is the stabilizer of the chosen base point in D. Choose the sign of χ_n so that

$$\mathcal{L}_n = \mathrm{SL}_2(\mathbb{R}) \times^{T_{\mathbb{R}}} \mathbb{C}_{\chi_n} \simeq \mathcal{O}_{\mathbb{P}^1}(-n)|_D.$$

The embedding and the line bundle can be pictured as follows:



The upper half-plane is an open orbit; its boundary, together with ∞, is $\mathbb{P}^1(\mathbb{R})$. Its complex structure is inherited from the complex flag variety, and the line bundle is a restriction:

$$\begin{array}{ccc} \mathcal{L}_n & \longhookrightarrow & \mathcal{O}_{\mathbb{P}^1}(-n) \\ \downarrow & & \downarrow \\ D & \xhookrightarrow{\text{open}} & \mathbb{P}^1(\mathbb{C}). \end{array}$$

This is the holomorphic counterpart of the smooth-section picture: we now use holomorphic sections and sheaf cohomology. The holomorphic and antiholomorphic discrete series at infinitesimal character ρ can be realized as

$$D_2^+ = \Gamma^{\mathrm{hol}}(D, \mathcal{L}_2)_{K_{\mathbb{R}}\text{-fin}}, \qquad D_2^- = H_c^1(D, \mathcal{L}_0)_{K_{\mathbb{R}}\text{-fin}}. \tag{2.2}$$

Here H_c^1 means compactly supported holomorphic sheaf cohomology.

2.3.1. *Computation for holomorphic discrete series.* Actually, $\mathbb{H}$ lies in a single coordinate chart of $\mathbb{P}^1(\mathbb{C})$, the line bundles $\mathcal{O}(n)$ are trivial on $\mathbb{H}$. For $\mathcal{O}(-1)$, it has a global section $e_{-1}:\mathbb{H}\to\mathcal{O}(-1)|_{\mathbb{H}}, z\mapsto(z,(z,1))$. So we have an isomorphism:

$$\Gamma^{\mathrm{hol}}(\mathbb{H},\mathcal{O}(-1))\simeq \mathrm{Hol}(\mathbb{H}),\quad f(z)e_{-1}(z)\mapsto f(z).$$

The $\mathrm{SL}_2(\mathbb{R})$ action on $\Gamma^{\mathrm{hol}}(\mathbb{H},\mathcal{O}(-1))$ is given by:

$$(g\cdot s)(z)=g(s(g^{-1}z))=(a-cz)^{-1}f(g^{-1}z)e_{-1}(z)$$

where $g=\begin{pmatrix}a & b\\ c & d\end{pmatrix}\in\mathrm{SL}_2(\mathbb{R})$ acts on $\mathbb{H}$ by Möbius transformations, $g^{-1}z=\dfrac{dz-b}{-cz+a}$.

For $\mathcal{O}(-2)$, we get a representation of $\mathrm{SL}_2(\mathbb{R})$ on $\mathrm{Hol}(\mathbb{H})$ by:

$$(g\cdot f)(z)=(a-cz)^{-2}f(g^{-1}z).$$

Moreover $\mathrm{Hol}(\mathbb{H})_{K\text{-fin}}=\oplus_{m=0}^{\infty}\mathbb{C}\cdot f_m$, where $f_m=\frac{(z-i)^m}{(z+i)^{m+2}}$ has K-type $2m+2$.

2.3.2. *Computation for anti-holomorphic discrete series.* Note that $H^1_c(\mathbb{H},\mathcal{O}(n))=\frac{A_c^{0,1}(\mathbb{H},\mathcal{O}(n))}{\bar\partial A_c^{0,0}(\mathbb{H},\mathcal{O}(n))}$.

Here $A_c^{0,1}(\mathbb{H},\mathcal{O}(n))=\{f(z)d\bar z\otimes e_n(z)\mid f\in C_c^\infty(\mathbb{H})\}$.

We have an isomorphism:

$$A_c^{0,1}(\mathbb{H},\mathcal{O}(n))\simeq C_c^\infty(\mathbb{H}),\quad y^nf(z)d\bar z\otimes e_n(z)\mapsto f(z).$$

The $\mathrm{SL}_2(\mathbb{R})$ action on $A_c^{0,1}(\mathbb{H},\mathcal{O}(n))$ is given by:

$$(g\cdot\alpha)(z)=(a-c\bar z)^{-n-2}f(g^{-1}z)\cdot y^nd\bar z\otimes e_n(z).$$

Where $\alpha(z)=y^nf(z)d\bar z\otimes e_n(z)$.

Therefore we have an action of $\mathrm{SL}_2(\mathbb{R})$ on $C_c^\infty(\mathbb{H})$ by:

$$(g\cdot f)(z)=(a-c\bar z)^{-n-2}f(g^{-1}z).$$

Passage to the quotient: $(H^1_c(\mathbb{H},\mathcal{O}(n)))_{K\text{-fin}}\simeq(\overline{\mathrm{Hol}}_{(2)}(\mathbb{H}))_{K\text{-fin}}$.

Moreover, for $\mathcal{O}(0)$, we have $(H^1_c(\mathbb{H},\mathcal{O}(0)))_{K\text{-fin}}\simeq\oplus_{m=0}^{\infty}\mathbb{C}\cdot f_{-m}$, where $f_{-m}=\frac{(\bar z-i)^m}{(\bar z+i)^{m+2}}$ has K-type $-2m-2$.

Remark 2.2. The Serre duality for Stein manifolds gives a natural isomorphism:

$$H^1_c(D,\mathcal{L}_0)\simeq\Gamma^{\mathrm{hol}}(D,\mathcal{L}_0^*\otimes\omega_D)^*=\Gamma^{\mathrm{hol}}(D,\mathcal{L}_2)^*.$$

This implies $D_2^-\simeq(D_2^+)^*$.

2.4. **Langlands Classification.**

Theorem 2.3. *At infinitesimal character ρ,*

$$\mathrm{Irr}_\rho(\mathrm{SL}_2(\mathbb{R}))=\{D_2^+,D_2^-,\mathbf{1},P_{1,1}\}.$$

There is a short exact sequence of Harish-Chandra modules

$$0\longrightarrow D_2^+\oplus D_2^-\longrightarrow P_{0,1}\longrightarrow\mathbf{1}\longrightarrow 0. \tag{2.3}$$

"Theorem" 2.4 (Langlands classification)**.** *Every irreducible admissible representation can be realized as a quotient of a standard module. A standard module is obtained by parabolic induction and cohomological induction from a character on a Cartan subgroup.*

Remark 2.5. Standard modules are analogue of Verma modules in the category $\mathcal{O}$ for semisimple Lie algebras.

At infinitesimal character ρ, the standard modules and their irreducible quotients for $\mathrm{SL}_2(\mathbb{R})$ are

Standard module	Irreducible quotient
$P_{0,1}$	$\mathbf{1}$
$P_{1,1}$	$P_{1,1}$
D_2^+	D_2^+
D_2^-	D_2^-

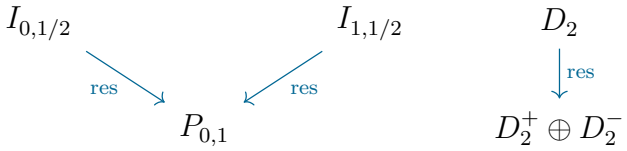
2.5. **Passing to** $\mathrm{PGL}_2(\mathbb{R})$**.** There are maps

$$\mathrm{SL}_2(\mathbb{R}) \longrightarrow\!\!\!\!\!\rightarrow \mathrm{PSL}_2(\mathbb{R}) \hookrightarrow \mathrm{PGL}_2(\mathbb{R}),$$

Only the representation $P_{1,1}$ does not descend through the first map, since $-I$ acts by -1.

There is an irreducible discrete series representation D_2 of the full group $\mathrm{PGL}_2(\mathbb{R})$ with

$$D_2|_{\mathrm{PSL}_2(\mathbb{R})} \simeq D_2^+ \oplus D_2^- .$$



The two principal series differ by the determinant-sign character, which becomes trivial on $\mathrm{PSL}_2(\mathbb{R})$. An element in the other component of $\mathrm{PGL}_2(\mathbb{R})$ interchanges the upper and lower half-planes and exchanges D_2^+ with D_2^-. Thus the full group joins the two discrete-series pieces.

The standard modules with infinitesimal ρ have the following quotients:

Standard module	Irreducible quotient
$I_{0,1/2}$	$\mathbf{1}$
$I_{1,1/2}$	sgn
D_2	D_2

Their composition series are expressed by

$$\begin{aligned} 0 &\longrightarrow D_2 \longrightarrow I_{0,1/2} \longrightarrow \mathbf{1} \longrightarrow 0, \\ 0 &\longrightarrow D_2 \longrightarrow I_{1,1/2} \longrightarrow \mathrm{sgn} \longrightarrow 0. \end{aligned} \tag{2.4}$$

3. Classification: enhanced Langlands correspondence

The focus of this section is the **classification problem**: parametrizing irreducible admissible representations. We introduce Langlands parameters and their enhancements, which distinguish representations within an L-packet and across the real forms in an inner class.

3.1. **Dual groups and the Weil group.** For a connected complex reductive group G, choose a maximal torus and a Borel to obtain the based root datum

$$\Psi(G) = (X^*(T), \Delta, X_*(T), \Delta^\vee).$$

The *dual group* ${}^\vee G$ is the connected complex reductive group whose based root datum is dual to $\Psi(G)$:

$$\Psi({}^\vee G) \simeq \Psi(G)^\vee.$$

Fix an inner class $[\sigma]$ of real forms. Its *L-group* is

$${}^L G = {}^\vee G \rtimes \Gamma, \qquad \Gamma = \mathrm{Gal}(\mathbb{C}/\mathbb{R}) = \{1, \gamma\},$$

where the action on ${}^\vee G$ is the pinned action determined by the inner class. Inner forms have the same L-group. For example,

$${}^L \mathrm{PGL}_2 = \mathrm{SL}_2(\mathbb{C}) \times \Gamma$$

for both $\mathrm{PGL}_2(\mathbb{R})$ and its compact inner form $\mathrm{SO}(3)$.

Definition 3.1. The *real Weil group* is

$$W_\mathbb{R} = \mathbb{C}^\times \sqcup \mathbb{C}^\times j, \qquad j^2 = -1, \qquad jzj^{-1} = \overline{z}.$$

Its quotient map $W_\mathbb{R} \to \Gamma$ sends $\mathbb{C}^\times$ to 1 and j to γ.

Definition 3.2. An *L-parameter* for the inner class is a continuous homomorphism

$$\varphi : W_\mathbb{R} \longrightarrow {}^L G$$

such that $\varphi(\mathbb{C}^\times)$ consists of semisimple elements and the diagram

$$\begin{array}{ccc} W_\mathbb{R} & \xrightarrow{\varphi} & {}^L G \\ & \searrow & \downarrow \\ & & \Gamma \end{array}$$

commutes. Write $P({}^L G)$ for the set of parameters and

$$\Phi({}^L G) = P({}^L G)/{}^\vee G$$

for conjugacy classes. In ABV terminology these are quasiadmissible parameters; for a fixed non-quasisplit real form, one must impose relevance to obtain its individual L-packets.

3.2. **Parameters as pairs (y, λ).**

Proposition 3.3. *There is a ${}^\vee G$-equivariant bijection*

$$P({}^L G) \simeq \left\{ (y, \lambda) \in ({}^L G \setminus {}^\vee G) \times {}^\vee \mathfrak{g} : \begin{array}{l} \lambda \text{ is semisimple}, \\ y^2 = \exp(2\pi i \lambda), \\ [\lambda, \mathrm{Ad}(y)\lambda] = 0 \end{array} \right\}.$$

ϕ corresponds to $(y(\phi), \lambda(\phi))$ where $\lambda = \lambda(\varphi)$ is the holomorphic exponent of $\varphi|_{\mathbb{C}^\times}$, and

$$y = y(\varphi) = \exp(\pi i \lambda)\varphi(j).$$

Definition 3.4. A *complete* or *enhanced L-parameter* is a pair (φ, τ), where $\varphi \in P({}^L G)$ and

$$\tau \in \operatorname{Irr} A_\varphi, \qquad A_\varphi = Z_{{}^\vee G}(\varphi)/Z_{{}^\vee G}(\varphi)^\circ.$$

For the following classification statement we assume G is adjoint. Write $\Phi^c({}^L G)$ for conjugacy classes of such enhanced parameters.

Theorem 3.5 (Enhanced real local Langlands correspondence)**.** *Suppose G is adjoint and fix the inner class $[\sigma]$, together with the normalization data for the correspondence. There is a bijection*

$$\operatorname{Irr}(G, [\sigma]) := \coprod_{[\sigma'] \in [\sigma]} \operatorname{Irr}\big(G(\mathbb{R}, \sigma')\big) \simeq \Phi^c({}^L G),$$

where the disjoint union is over equivalence classes of real forms in that inner class.

Remark 3.6. The normalizetion data include a choice of Whittaker datum on $G(\mathbb{R}, \sigma)$ and a choice of distinguished involution on the dual group ${}^\vee G$. Here the group G is adjoint so the choice of Whittaker datum is unique up to conjugation. And the distinguished involution we choose the standard one ${}^\vee G \cdot I\gamma$.

For non-adjoint G, the general ABV formulation uses strong real forms and component groups in an appropriate covering of the dual group.

3.3. **The example $G = \mathrm{PGL}_2(\mathbb{C})$.** Here ${}^\vee G = \mathrm{SL}_2(\mathbb{C})$, and we identify the infinitesimal character with

$$\rho = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \in {}^\vee\mathfrak{g}.$$

All three parameter classes below have a representative with $\lambda = \rho$. γ is the nontrivial element of Γ and acts on $\mathrm{SL}_2(\mathbb{C})$ trivially. The parameters and corresponding representations of inner forms are listed below.

$\mathrm{PGL}_2(\mathbb{R})$	$\mathrm{SO}(3)$	y	$Z_{{}^\vee G}(y, \rho)$
1	—	$\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \gamma$	${}^\vee T$
sgn	—	$\begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} \gamma$	${}^\vee T$
D_2	**1**	$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \gamma$	$\{\pm I\}$

4. Structure: geometric parameters

We now turn to the **structure problem**: studying the characters of irreducible representations and their multiplicities in standard representations. ABV's idea is to modify the space of Langlands parameters to some space of geometric parameters, whose orbits closure relations are more interesting.

4.1. **Canonical flats and the parameter variety.** Let $\lambda \in {}^\vee\mathfrak{g}$ be semisimple. Define

$$\begin{aligned} e(\lambda) &= \exp(2\pi i\lambda), & M(\lambda) &= Z_{{}^\vee G}(e(\lambda)),\\ \mathfrak{m}(\lambda) &= \operatorname{Lie} M(\lambda) = \bigoplus_{n\in\mathbb{Z}} \mathfrak{m}(\lambda)_n, & \mathfrak{m}(\lambda)_n &= \{X \in {}^\vee\mathfrak{g} : [\lambda, X] = nX\},\\ \mathfrak{l}(\lambda) &= \mathfrak{m}(\lambda)_0, & \mathfrak{u}(\lambda) &= \bigoplus_{n\geq 1} \mathfrak{m}(\lambda)_n,\\ \mathfrak{p}(\lambda) &= \mathfrak{l}(\lambda) \oplus \mathfrak{u}(\lambda). \end{aligned}$$

The sum uses only the integral eigenvalues of $\operatorname{ad}(\lambda)$. Let $L(\lambda) = Z_{{}^\vee G}(\lambda)$, let $U(\lambda)$ be the connected unipotent subgroup with Lie algebra $\mathfrak{u}(\lambda)$, and put

$$P(\lambda) = L(\lambda)U(\lambda) \subset M(\lambda).$$

This is a parabolic subgroup of $M(\lambda)$.

Definition 4.1. The *canonical flat* through λ is

$$F(\lambda) = \lambda + \mathfrak{u}(\lambda).$$

It lies in the adjoint orbit of λ, and the exponential $\exp(2\pi i\,\cdot)$ is constant on it. Write its common value as $e(F(\lambda)) = e(\lambda)$.

Definition 4.2. A *geometric parameter* is a pair (y, F) with

$$y \in {}^L G \setminus {}^\vee G, \qquad F \text{ a canonical flat}, \qquad y^2 = e(F).$$

Write $X({}^L G)$ for the set of geometric parameters, and

$$\Xi({}^L G) = X({}^L G)/{}^\vee G$$

for its set of orbits.

To obtain an algebraic variety, fix a semisimple adjoint orbit $\mathcal{O} \subset {}^\vee\mathfrak{g}$, and let

$$X(\mathcal{O}, {}^L G) = \{(y, F) \in X({}^L G) : F \subset \mathcal{O}\}.$$

Set

$$\begin{aligned} \mathcal{F}(\mathcal{O}) &= \{F : F \text{ is a canonical flat contained in } \mathcal{O}\},\\ e(\mathcal{O}) &= \{\exp(2\pi i\lambda) : \lambda \in \mathcal{O}\},\\ \sqrt{e(\mathcal{O})} &= \{y \in {}^L G \setminus {}^\vee G : y^2 \in e(\mathcal{O})\}. \end{aligned}$$

For a base point $\lambda \in \mathcal{O}$,

$$\mathcal{F}(\mathcal{O}) \simeq {}^\vee G/P(\lambda), \qquad e(\mathcal{O}) \simeq {}^\vee G/M(\lambda).$$

The space $\sqrt{e(\mathcal{O})}$ has finitely many ${}^\vee G$-orbits. Choose representatives y_i with $y_i^2 = e(\lambda)$, and put

$$K_i = Z_{{}^\vee G}(y_i) \subset M(\lambda).$$

On $M(\lambda)$, conjugation by y_i is an involution, with fixed group K_i. The corresponding pieces of the parameter variety are

$$\begin{aligned} X_i(\mathcal{O}, {}^L G) &\simeq {}^\vee G/P(\lambda) \times_{{}^\vee G/M(\lambda)} {}^\vee G/K_i\\ &\simeq {}^\vee G \times^{K_i} \big(M(\lambda)/P(\lambda)\big). \end{aligned} \tag{4.1}$$

Once the square-root orbit is fixed, the remaining geometry is the K_i-action on the flag variety $M(\lambda)/P(\lambda)$; the factor ${}^\vee G$ transports this fiber over ${}^\vee G/K_i$. Thus $X(\mathcal{O}, {}^L G)$ is a smooth complex algebraic variety with finitely many ${}^\vee G$-orbits. The unrestricted set is a disjoint union

$$X({}^L G) = \coprod_{\mathcal{O}} X(\mathcal{O}, {}^L G);$$

4.2. **Complete geometric parameters and local systems.**

Theorem 4.3. *The ${}^\vee G$-equivariant map*

$$P({}^L G) \longrightarrow X({}^L G), \qquad (y, \lambda) \longmapsto (y, F(\lambda))$$

induces a bijection

$$\Phi({}^L G) \simeq \Xi({}^L G).$$

It also induces an isomorphism of component groups

$$\frac{Z_{{}^\vee G}(y, \lambda)}{Z_{{}^\vee G}(y, \lambda)^\circ} \simeq \frac{Z_{{}^\vee G}(y, F(\lambda))}{(Z_{{}^\vee G}(y, F(\lambda)))^\circ}.$$

Here ${}^\vee G^{(y,F)}$ is the stabilizer of the pair, so it preserves F setwise, rather than pointwise.

Continue with the adjoint assumption of Theorem 3.5. A *complete geometric parameter* is a conjugacy class of pairs (ξ, τ), where

$$\xi \in X({}^L G), \qquad \tau \in \mathrm{Irr}\big(Z_{{}^\vee G}(\xi)/(Z_{{}^\vee G}(\xi))^\circ\big).$$

Equivalently, it is a pair $(S, \mathcal{V})$ consisting of a ${}^\vee G$-orbit $S \subset X({}^L G)$ and an irreducible ${}^\vee G$-equivariant local system $\mathcal{V}$ on S. Denote the set of these parameters by $\Xi^c({}^L G)$.

$$\Phi^c({}^L G) \xleftrightarrow{\text{1:1}} \Xi^c({}^L G)$$

$$[(\varphi, \tau)] \longmapsto (S, \mathcal{V})$$

Fix $X = X(\mathcal{O}, {}^L G)$ and write $j_S : S \hookrightarrow X$ for an orbit inclusion. With the usual cohomological shift convention, put

$$\mu(S, \mathcal{V}) = (j_S)_! \mathcal{V}[\dim S], \qquad p(S, \mathcal{V}) = (j_S)_{!*} \mathcal{V}[\dim S]. \tag{4.2}$$

The intermediate extension $p(S, \mathcal{V})$ is a simple equivariant perverse sheaf. In general, however, $\mu(S, \mathcal{V})$ is a constructible complex and need not be perverse. Its class in the Grothendieck group of equivariant perverse sheaves is the perverse Euler characteristic:

$$[\mu(S, \mathcal{V})] = \sum_{k \in \mathbb{Z}} (-1)^k \big[{}^p H^k(\mu(S, \mathcal{V}))\big] \quad \text{in } K_0\big(\mathrm{Perv}_{{}^\vee G}(X)\big).$$

This is an equality of classes, rather than a direct-sum decomposition; taking only ${}^p H^0$ would in general lose the other contributions.

For the regular case used in this talk, $M(\lambda)/P(\lambda)$ is a full flag variety, and every K_i-orbit inclusion is affine. By the associated-bundle description (4.1), the same holds for the ${}^\vee G$-orbit inclusions j_S. Both $(j_S)_!$ and $R(j_S)_*$ are then exact for the perverse t-structure, so $\mu(S, \mathcal{V})$ and the costandard extension $R(j_S)_* \mathcal{V}[\dim S]$ are perverse. Thus ${}^p H^k(\mu(S, \mathcal{V})) = 0$ for $k \neq 0$.

4.3. **The PGL_2 example as geometry on $\mathbb{P}^1$.** Take again

$$G = \mathrm{PGL}_2(\mathbb{C}), \qquad {}^\vee G = \mathrm{SL}_2(\mathbb{C}), \qquad \mathcal{O} = \mathrm{Ad}({}^\vee G)\rho.$$

Since $\exp(2\pi i\rho) = -I$,

$$M(\rho) = {}^\vee G, \qquad P(\rho) = {}^\vee B, \qquad e(\mathcal{O}) = \{-I\}.$$

There is one conjugacy class of square roots of $-I$ in ${}^L G \setminus {}^\vee G$, with representative

$$y_0 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \gamma, \qquad Z_{{}^\vee G}(y_0) = {}^\vee T.$$

Consequently,

$$X(\mathcal{O}, {}^L G) \simeq {}^\vee G \times^{{}^\vee T} ({}^\vee G / {}^\vee B).$$

Equivariant perverse sheaves on this associated bundle can be studied on ${}^\vee G/{}^\vee B \simeq \mathbb{P}^1$:

$$\mathrm{Perv}_{{}^\vee G}\big(X(\mathcal{O}, {}^L G)\big) \simeq \mathrm{Perv}_{{}^\vee T}(\mathbb{P}^1).$$

The equivalence uses smooth pullback with relative dimension shift [2], so the following descriptions use the dimensions of orbits on $\mathbb{P}^1$.

Write ${}^\vee T = \{\mathrm{diag}(t, t^{-1}) : t \in \mathbb{C}^\times\}$. Its action on the affine coordinate is $z \mapsto t^2 z$. There are three orbits:

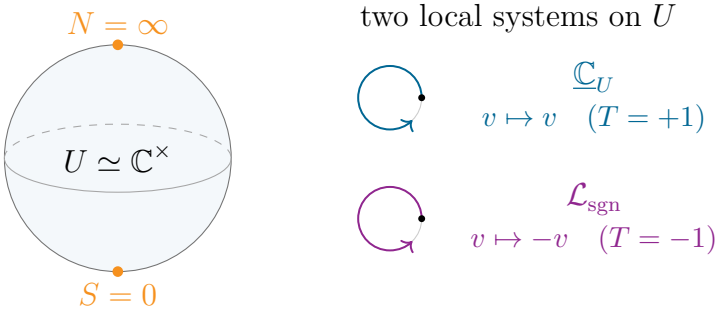
$$N = \{\infty\}, \qquad S = \{0\}, \qquad U = \mathbb{P}^1 \setminus \{N, S\} \simeq \mathbb{C}^\times.$$

The point stabilizers at N and S are connected. On U the stabilizer is $\{\pm I\}$, giving two irreducible equivariant local systems:

$$\underline{\mathbb{C}}_U \qquad \text{and} \qquad \mathcal{L}_{\mathrm{sgn}}.$$

Their monodromies are $+1$ and -1, respectively. Let $j : U \hookrightarrow \mathbb{P}^1$ and i_N, i_S be the inclusions of the two points, and set

$$\delta_N = (i_N)_* \mathbb{C}, \qquad \delta_S = (i_S)_* \mathbb{C}.$$



The four complete geometric parameters give the following objects. The labels N and S are chosen to match the normalization above.

$(S, \mathcal{V})$	$M(S, \mathcal{V})$	$\pi(S, \mathcal{V})$	$\mu(S, \mathcal{V})$	$p(S, \mathcal{V})$
$(N, \mathbb{C})$	$I_{0,1/2}$	$\mathbf{1}_{\mathrm{PGL}_2(\mathbb{R})}$	δ_N	δ_N
$(S, \mathbb{C})$	$I_{1,1/2}$	$\mathrm{sgn}_{\mathrm{PGL}_2(\mathbb{R})}$	δ_S	δ_S
$(U, \underline{\mathbb{C}}_U)$	D_2	D_2	$j_! \underline{\mathbb{C}}_U[1]$	$\underline{\mathbb{C}}_{\mathbb{P}^1}[1]$
$(U, \mathcal{L}_{\mathrm{sgn}})$	$\mathbf{1}_{\mathrm{SO}(3)}$	$\mathbf{1}_{\mathrm{SO}(3)}$	$j_! \mathcal{L}_{\mathrm{sgn}}[1]$	$j_{!*} \mathcal{L}_{\mathrm{sgn}}[1]$

Here $M(S, \mathcal{V})$ is the standard module and $\pi(S, \mathcal{V})$ is its irreducible quotient.

4.4. **An exercise with perverse sheaves.**

Lemma 4.4. *Let $\mathcal{L}$ be a ${}^\vee T$-equivariant local system on U, corresponding to a representation V of $\mathbb{Z}/2\mathbb{Z}$. Write $V^{\mathbb{Z}/2\mathbb{Z}}$ for invariants and $V_{\mathbb{Z}/2\mathbb{Z}}$ for coinvariants. Show that there is an exact sequence in* $\mathrm{Perv}_{{}^\vee T}(\mathbb{P}^1)$*:*

$$\begin{aligned} 0 &\longrightarrow (j_N)_* V^{\mathbb{Z}/2\mathbb{Z}} \oplus (j_S)_* V^{\mathbb{Z}/2\mathbb{Z}} \\ &\longrightarrow j_!\mathcal{L}[1] \longrightarrow Rj_*\mathcal{L}[1] \\ &\longrightarrow (j_N)_* V_{\mathbb{Z}/2\mathbb{Z}} \oplus (j_S)_* V_{\mathbb{Z}/2\mathbb{Z}} \longrightarrow 0. \end{aligned} \tag{4.3}$$

The image of the middle map is $j_{!*}\mathcal{L}[1]$.

Proof. Exercise. □

For the trivial local system, the sequence becomes

$$\begin{aligned} 0 &\longrightarrow \delta_N \oplus \delta_S \longrightarrow j_!\underline{\mathbb{C}}_U[1] \\ &\longrightarrow Rj_*\underline{\mathbb{C}}_U[1] \longrightarrow \delta_N \oplus \delta_S \longrightarrow 0, \end{aligned} \tag{4.4}$$

with middle image $\underline{\mathbb{C}}_{\mathbb{P}^1}[1]$. In particular,

$$0 \longrightarrow \delta_N \oplus \delta_S \longrightarrow j_!\underline{\mathbb{C}}_U[1] \longrightarrow \underline{\mathbb{C}}_{\mathbb{P}^1}[1] \longrightarrow 0.$$

For $\mathcal{L}_{\mathrm{sgn}}$, both invariants and coinvariants vanish, so

$$j_!\mathcal{L}_{\mathrm{sgn}}[1] \simeq j_{!*}\mathcal{L}_{\mathrm{sgn}}[1] \simeq Rj_*\mathcal{L}_{\mathrm{sgn}}[1].$$

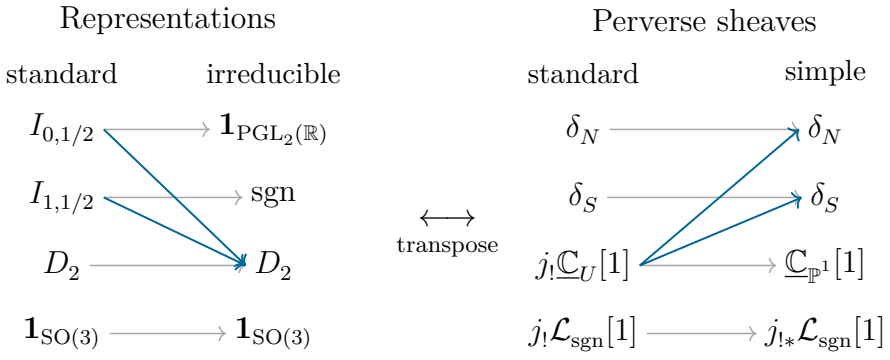
4.5. **Multiplicity matrices and Vogan duality.** All equalities in this subsection are in the relevant Grothendieck groups. The representation-theoretic exact sequences give

$$\begin{pmatrix} [I_{0,1/2}] \\ [I_{1,1/2}] \\ [D_2] \\ [\mathbf{1}_{\mathrm{SO}(3)}] \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} [\mathbf{1}_{\mathrm{PGL}_2(\mathbb{R})}] \\ [\mathrm{sgn}] \\ [D_2] \\ [\mathbf{1}_{\mathrm{SO}(3)}] \end{pmatrix}. \tag{4.5}$$

The sheaf-theoretic exact sequences give

$$\begin{pmatrix} [\delta_N] \\ [\delta_S] \\ [j_!\underline{\mathbb{C}}_U[1]] \\ [j_!\mathcal{L}_{\mathrm{sgn}}[1]] \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} [\delta_N] \\ [\delta_S] \\ [\underline{\mathbb{C}}_{\mathbb{P}^1}[1]] \\ [j_{!*}\mathcal{L}_{\mathrm{sgn}}[1]] \end{pmatrix}. \tag{4.6}$$

The following diagrams display the composition factors: each arrow contributes one to the corresponding matrix entry. The extra factors are highlighted.



The two extra arrows into D_2 on the representation side become two extra arrows out of $j_!\underline{\mathbb{C}}_U[1]$ on the sheaf side: the factors δ_N and δ_S are exactly the boundary contributions seen above. Thus, in this example and with the displayed ordering, the two multiplicity matrices are transposes of each other. This illustrates the relation underlying Vogan duality.

References

[1] J. Adams, D. Barbasch, and D. A. Vogan, Jr., *The Langlands Classification and Irreducible Characters for Real Reductive Groups*, Progress in Mathematics 104, Birkhäuser, 1992. Author-hosted text.

[2] L. Mason-Brown, *Arthur's Conjectures and the Orbit Method for Real Reductive Groups*, arXiv:2204.04994 [math.RT], 2022.